Scan4Disease

Presentation & Testing Demo Guide (Mobile App & Clinician Web Portal)

🔑 1. Demo Logins & Access Directory

Platform	Role	User Name	Email ID	Password	Capabilities & Notes
Web Portal `/portal/login`	**Doctor 1 (Primary)**	Dr. A. Rao	`dr.rao@example.com`	`Str0ngPass!`	Reg: MH-12345 · Chief Dermatologist. Full patient roster & reviews.
Web Portal `/portal/login`	**Doctor 2 (Clinician)**	Dr. Sunita Mehta	`dr.mehta@example.com`	`Str0ngPass!`	Reg: KA-67890 · Consultant Dermatologist & Dermatosurgeon.
Web Portal `/portal/login`	**Doctor 3 (Clinician)**	Dr. Vikram Kapoor	`dr.kapoor@example.com`	`Str0ngPass!`	Reg: DL-98765 · Pediatric & Aesthetic Dermatology.
Web Portal `/portal/login`	**Doctor 4 (Clinician)**	Dr. Priya Nambiar	`dr.nambiar@example.com`	`Str0ngPass!`	Reg: KL-45678 · Clinical Dermatopathology Specialist.
Web Portal `/portal/login`	**Doctor 5 (Clinician)**	Dr. Rajesh Deshmukh	`dr.deshmukh@example.com`	`Str0ngPass!`	Reg: MH-54321 · Senior Cutaneous Oncology Consultant.
Web Portal `/portal/login`	**Doctor 6 (Clinician)**	Dr. Ananya Sen	`dr.sen@example.com`	`Str0ngPass!`	Reg: WB-34567 · Melanoma & Pigmentary Lesion Specialist.

Platform	Role	User Name	Email ID	Password	Capabilities & Notes
Mobile App `/auth/login`	**Patient 1 (Primary)**	Raj	`raj@gmail.com`	`12345678`	Preloaded history: Melanocytic nevus & Basal cell carcinoma.
Mobile App `/auth/login`	**Patient 2 (Patient)**	Ananya Verma	`ananya@gmail.com`	`12345678`	Preloaded history: Actinic keratosis & Benign keratosis.
Mobile App `/auth/login`	**Demo Patients**	Priya / Sam / Jordan	`priya@example.com` `sam@example.com`	`12345678`	Clinical demo cases linked to doctors in worklists.

📸 2. Five Curated Test Photos Across Lesion Criteria

All photos are located in the local workspace directory: `demo_test_samples/`

🔴 Photo 1: Malignant Melanoma (High Risk)

URGENT MEDICAL EVALUATION

File Location: `demo_test_samples/01_melanoma_malignant.jpg`

Diagnostic Criteria (ABCDE): Asymmetric borders, dark pigmented irregular margins, variegation, diameter >6mm, fast evolution.

Symptom Inputs: Bleeding = *Yes*, Itching = *Yes*, Duration = *8 weeks*.

Expected Model Output: Predicted: `Melanoma (mel)` · Confidence: `75.6%` · Triage: `Urgent medical evaluation` · Grad-CAM: Saliency centered on core pigment.

Structured AI Assistant Questions:

- "*What is the clinical significance of irregular borders in this lesion?*"
- "*Why does the app recommend urgent medical evaluation?*"

🟠 Photo 2: Basal Cell Carcinoma (Malignant)

PROMPT CONSULTATION

File Location: `demo_test_samples/02_basal_cell_carcinoma.jpg`

Diagnostic Criteria: Pearly nodule, translucent border, visible telangiectasia (tiny vessels), central indentation.

Symptom Inputs: Bleeding = *Occasional*, Pain = *Mild*, Duration = *20 weeks*.

Expected Model Output: Predicted: `Basal cell carcinoma (bcc)` · Confidence: `77.7%` · Triage: `Urgent / Prompt medical evaluation`.

Structured AI Assistant Questions:

- "*What are standard treatment options for BCC (e.g. Mohs surgery)?*"
- "*Does basal cell carcinoma usually metastasize?*"

Photo 3: Actinic Keratosis (Pre-malignant)

PROMPT CONSULTATION

File Location: `demo_test_samples/03_actinic_keratosis_precancer.jpg`

Diagnostic Criteria: Erythematous, rough, sandpaper-like scaly patch on chronic sun-exposed skin area.

Symptom Inputs: Sun exposure = *High*, Rough surface = *Yes*, Duration = *12 weeks*.

Expected Model Output: Predicted: `Actinic keratosis (akiec)` · Confidence: `89.3%` · Triage: `Prompt dermatologist consultation`.

Structured AI Assistant Questions:

- "*Can actinic keratosis progress into invasive squamous cell carcinoma?*"
- "*What preventive care and sun protection steps should be taken?*"

Photo 4: Melanocytic Nevus (Benign Mole)

ROUTINE CONSULTATION

File Location: `demo_test_samples/04_benign_melanocytic_nevus.jpg`

Diagnostic Criteria: Symmetrical round shape, uniform pigment network, distinct smooth borders, stable long-term history.

Symptom Inputs: Bleeding = *No*, Itching = *No*, Duration = *> 1 year (Stable)*.

Expected Model Output: Predicted: `Mole (nv)` · Confidence: `83.4%` · Triage: `Routine dermatologist consultation`.

Structured AI Assistant Questions:

- "*What signs indicate that a normal mole might be turning malignant?*"

- *"How frequently should routine skin checks be performed?"*

🛡️ Photo 5: Skin Abrasion / Wound (Safety Gate & Router)

NON-LESION SAFETY GATE TRIGGERED

File Location: `demo_test_samples/05_skin_abrasion_wound.jpg`

Diagnostic Criteria: Superficial mechanical scratch / wound / non-lesion damage. Multi-stage safety router halts classification before disease mapping.

Expected Model Output: Pipeline Outcome: `OTHER_DAMAGE` · Router Confidence: `99.3% Non-Lesion` · Triage: First-aid and hygiene guidance.

Structured AI Assistant Questions:

- *"What are the best first-aid practices for a minor skin scratch or abrasion?"*
- *"How can I tell if a skin wound has become infected?"*

🚀 3. Step-by-Step Walkthrough Flow

1. **Start Backend:** Run `powershell -ExecutionPolicy Bypass -File scripts\run_backend.ps1`
2. **Connect Phone:** Run `powershell -ExecutionPolicy Bypass -File scripts\connect_phone.ps1`
3. **Patient Screening (App):** Sign in as `raj@gmail.com` / `12345678` → Scan photo → View AI report → Tap "*Share with a Doctor*" → Select *Dr. A. Rao*.
4. **Doctor Review (Web Portal):** Open `http://localhost:8000/portal/login` → Sign in as `dr.rao@example.com` / `Str0ngPass!` → Open Patient Worklist → Inspect photo, Grad-CAM, & answers → Enter doctor note and update review status.

💡 4. Pro-Tips & Diagnostic Commands

Diagnostic Action	Command
Re-connect Phone Tunnel	`powershell -ExecutionPolicy Bypass -File scripts\connect_phone.ps1`
Verify All Logins	`.\.venv\Scripts\python.exe backend\tests\verify_logins.py`
Pre-Demo Smoke Test	`.\.venv\Scripts\python.exe scripts\demo_check.py`

Diagnostic Action	Command
Re-seed Demo Database	`.\.venv\Scripts\python.exe scripts\seed_portal_demo.py`